Basic Hydrologic Concepts

Katie Markovich, EPS 522/ ENVS 423L (Fall 2025)

Primary Dimensions

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8. Angle, θ, with SI units of radians (rad) or degrees ($^\circ$)

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5. Density, ρ, (M/V)

Dimensional Analysis

Analysis of relationships between different physical quantities by identifying their base dimensions and units. A fancy way of saying, check the units!

What are the dimensions of Force and Pressure?

State Variables

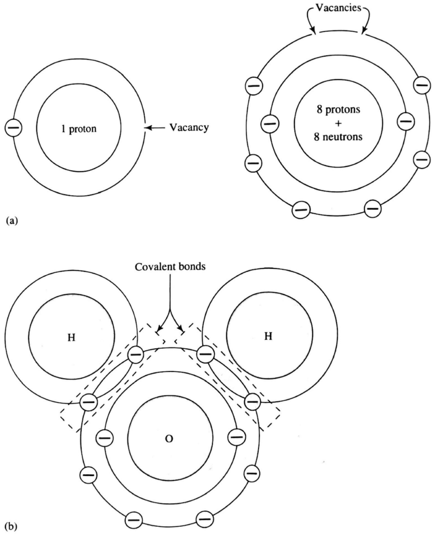
State variables describe a condition of a system at a specific time and location.
Examples include:

-
-
-
-
-
-
-

Water Molecule

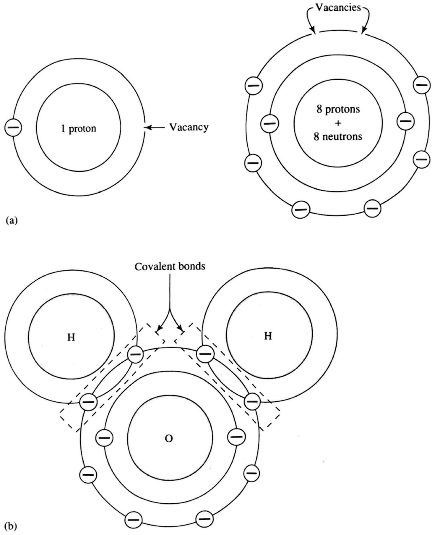
The molecular formula of water is H_2O, with two hydrogen atoms and one oxygen atom. The outer shell of hydrogen has 1 electron but can accommodate 2 and the outer shell of oxygen has 6 electrons but can accommodate 8.

This mutual sharing of electrons is a **covalent bond**.



Features of Water Molecule

- Covalent bonds are very strong (much energy needed to break them)



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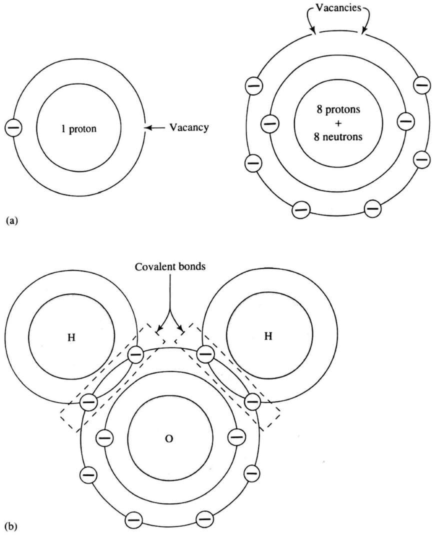

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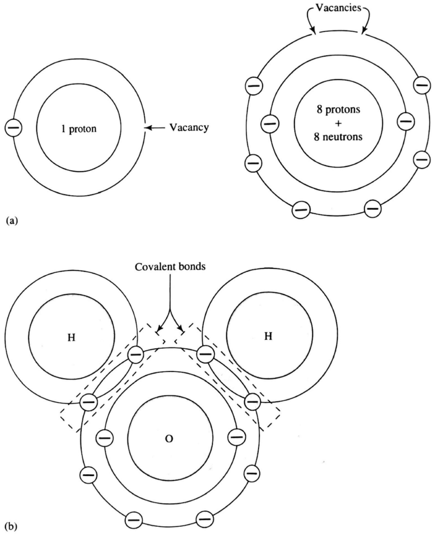

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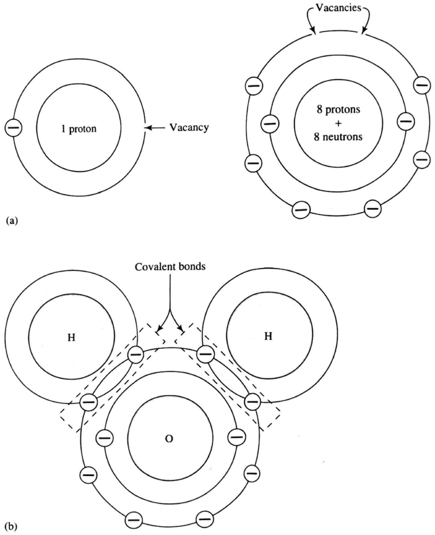

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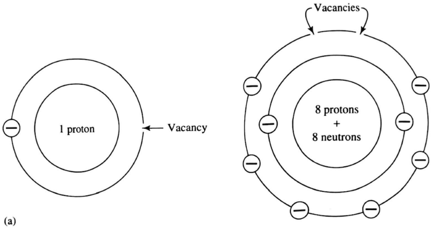
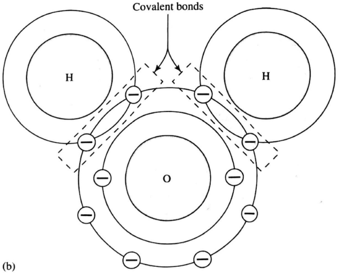
- These features explain much of the unusual properties of water.





Heat Capacity

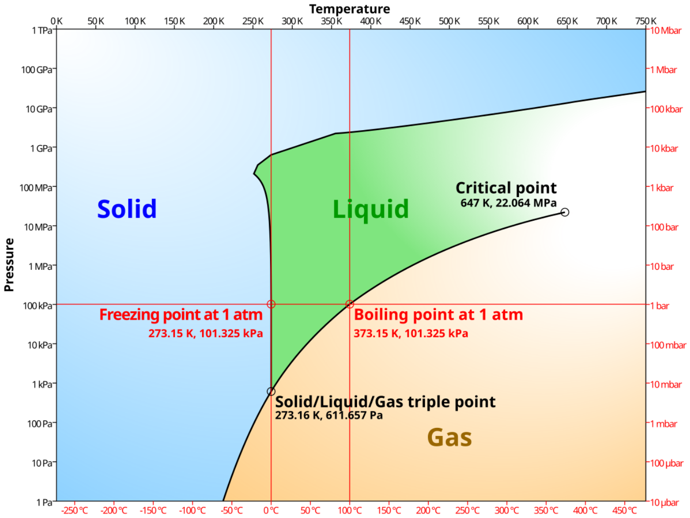
The strong hydrogen bonds of the water molecule lead to a high heat capacity, or the energy required to raise one gram of water 1 °C

Substance	c_p (J/kg°K)*	ρ (kg/m^3)
Freshwater (liquid)	4.186×10^3	1.00×10^3
Freshwater (ice)	2.108×10^3 (at 0 °C)	0.92×10^3
Seawater	3.93×10^3 (at 2.2 °C)	1.025×10^3
Air	1.005×10^3	1.205 (at 20 °C)
Granite	7.9×10^2	**2.75×10^3
Marble	8.8×10^2	**2.80×10^3
Sandstone	9.2×10^2	**2.80×10^3
Limestone	8.4×10^2	**2.70×10^3
Dolostone	9.2×10^2	**2.90×10^3
Clay	9.2×10^2	*1.8×10^3 - 2.6×10^3
Quartz	7.1×10^2 (at 0 °C)	**2.64×10^3
Calcite	8.4×10^2 (average over 0–100 °C)	**2.71×10^3

*Engineering Tool Box (2007).

**Cote and Konrad (2005).

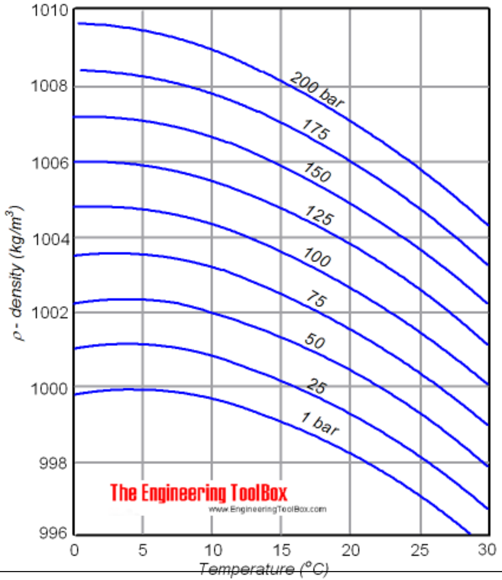
Phase Diagram



Density

$$\rho = \frac{m}{V}$$

where ρ is density, m is mass (in units M), and V is volume (in units L^3).



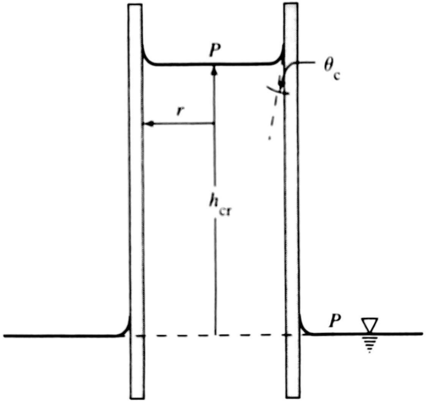
Surface Tension

Hydrogen bonds produce a net inward force to molecules at the surface of water called surface tension (σ) having units of force (F) divided by distance over which it acts (L^{-1})

σ is equal to 0.0756 Nm^{-1} at 0°C but decreases with increasing temperature or in presence of dissolved surfactants.

Capillary Rise

The polarity of water also leads to an attraction of molecules to surfaces, leading to water being drawn upward where flow scale is less than a few millimeters (i.e., pores).



$$F_u = \sigma \cdot \cos(\theta_c) \cdot 2 \cdot \pi \cdot r$$

$$F_d = \gamma \cdot \pi \cdot r^2 \cdot h_{cr}$$

Where, σ is surface tension, θ_c is contact

angle, r is radius of cylinder, γ is weight density of water, and h_{cr} is height to which water will rise.

Solving for Capillary Rise

Since Newton's 3rd Law of Motion states that for every action force, there is an equal and opposite reaction force, we can combine the equations of upward and downward force to solve for capillary rise (h_{cr}).

We know (or can estimate) σ and γ, and we can measure θ_c and r, we can solve for the expected capillary rise in a cylinder.

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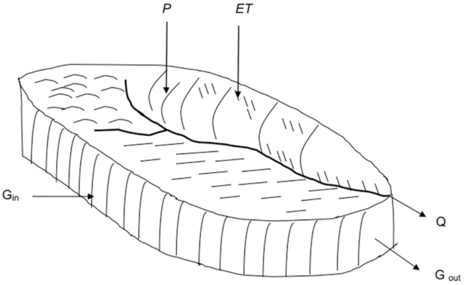
- Fick's First Law of Diffusion, a substance moves from higher to lower concentrations at a rate that is proportional to the concentration gradient.

Conservation Equations

Amount In - Amount Out = Change in Storage

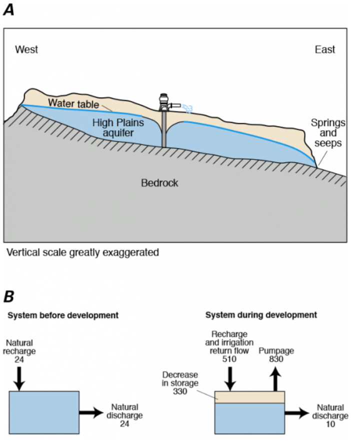
for a defined control volume, a defined period of time, and a conservative substance.

Control Volume



Where:

P	=	Precipitation
ET	=	Evapotranspiration
Q	=	Stream outflow
G_{in}	=	Ground water in
G_{out}	=	Ground water out

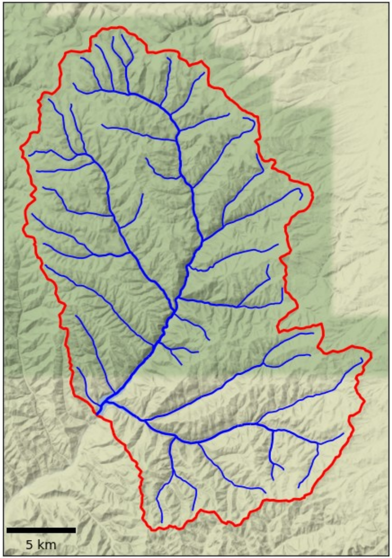


Does not have to be physically contiguous (i.e., glaciers)

Watershed Delineation

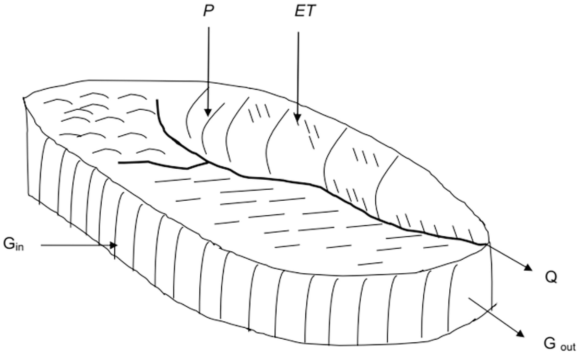
Watershed is most common control volume, defined as area that appears (on basis of topography) to contribute all water that passes through a given cross-section of a stream.

Nowadays, it is trivial to delineate a watershed based on an arbitrary point along a stream using a Digital Elevation Model (DEM) and openly available GIS tools.



Regional Water Balance

$$P + G_{in} - (Q + ET + G_{out}) = \Delta S$$



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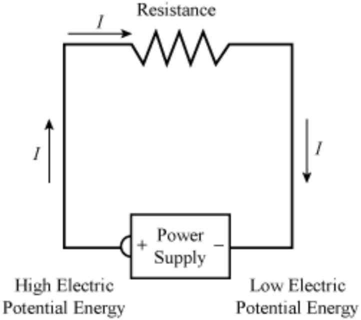
Challenges to Estimating Regional Water Balance

Electrical Circuit Analog

The flow of water can often be conceptualized as analogous to an electric current, where water flows from high to low “potential”, modulated by some form of "resistance"



Conductance is the inverse of resistance, and describes how easily water is conducted. In groundwater, conductance is calculated as:

$$C = \frac{KA}{L}$$

where K is hydraulic conductivity, A is area, and L is distance.

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Deliverables: technical memorandum and presentation to the class.

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4. Identify the primary water resource management concern(s) (e.g., climate change, land use change, contamination, drought, etc.) and outline potential next steps.

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